

WEB DEVELOPMENT

SE102.3

CHAPTER 05

PHP – Part C

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PHP Database Connection

Connecting to a MySQL database using PHP involves creating a connection object with the required credentials such as server name, username, password, and database name.

The connection is established using the `mysqli` extension, which provides the necessary functions to interact with MySQL databases.

Once connected, you can execute queries to retrieve or manipulate data.

Proper error handling is essential to manage connection issues and ensure a smooth database interaction.

```
<?php
// Database configuration
$servername = "localhost";
$username = "root"; // Default XAMPP username
$password = ""; // Default XAMPP password (empty)
$dbname = "webmodule1"; // Replace with your database name

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error){
    echo "Connection Failed";
}
else {
    echo "Connected Successfully";
}

// Close connection
$conn->close();
?>
```

Create a MySQL Table Using MySQLi

```
// SQL statement to create a table
$sql = "CREATE TABLE teacher (
    id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    firstname VARCHAR(30) NOT NULL,
    lastname VARCHAR(30) NOT NULL,
    email VARCHAR(50)
)";

// Execute the query and check for errors
if ($conn->query($sql) === TRUE) {
    echo "Teacher table created successfully";
} else {
    echo "Error creating table: " . $conn->error;
}
```

Insert Data Into MySQL Using MySQLi

```
// SQL statement to insert data
$sql_insert = "INSERT INTO teacher (firstname, lastname, email) VALUES
    ('John', 'Doe', 'john.doe@example.com'),
    ('Jane', 'Smith', 'jane.smith@example.com')";

// Execute the query and check for errors
if ($conn->query($sql_insert) === TRUE) {
    echo "New records created successfully<br>";
} else {
    echo "Error inserting data: " . $conn->error . "<br>";
}
```

Select Data From a MySQL Database

```
// SQL statement to select data
$sql_select = "SELECT id, firstname, lastname, email FROM staff";

// Execute the query and store the result
$result = $conn->query($sql_select);
```

```

if ($result === FALSE) {
    echo "Error executing query: " . $conn->error . "<br>";
} else if ($result->num_rows > 0) {
    // Output data in an HTML table
    echo "<table border='1'>
        <tr>
            <th>ID</th>
            <th>First Name</th>
            <th>Last Name</th>
            <th>Email</th>
        </tr>";

    // Fetch each row of data from the result set
    while($row = $result->fetch_assoc()) {
        echo "<tr>
            <td>" . $row["id"] . "</td>
            <td>" . $row["firstname"] . "</td>
            <td>" . $row["lastname"] . "</td>
            <td>" . $row["email"] . "</td>
        </tr>";
    }
    echo "</table>";
} else {
    echo "0 results";
}

```

New records created successfully

ID	First Name	Last Name	Email
1	John	Doe	john.doe@example.com
2	Jane	Smith	jane.smith@example.com
3	John	Doe	john.doe@example.com
4	Jane	Smith	jane.smith@example.com

Delete record from a MySQL Database

```
// Check if a delete request has been made
if (isset($_POST['delete_id'])) {
    $delete_id = $_POST['delete_id'];

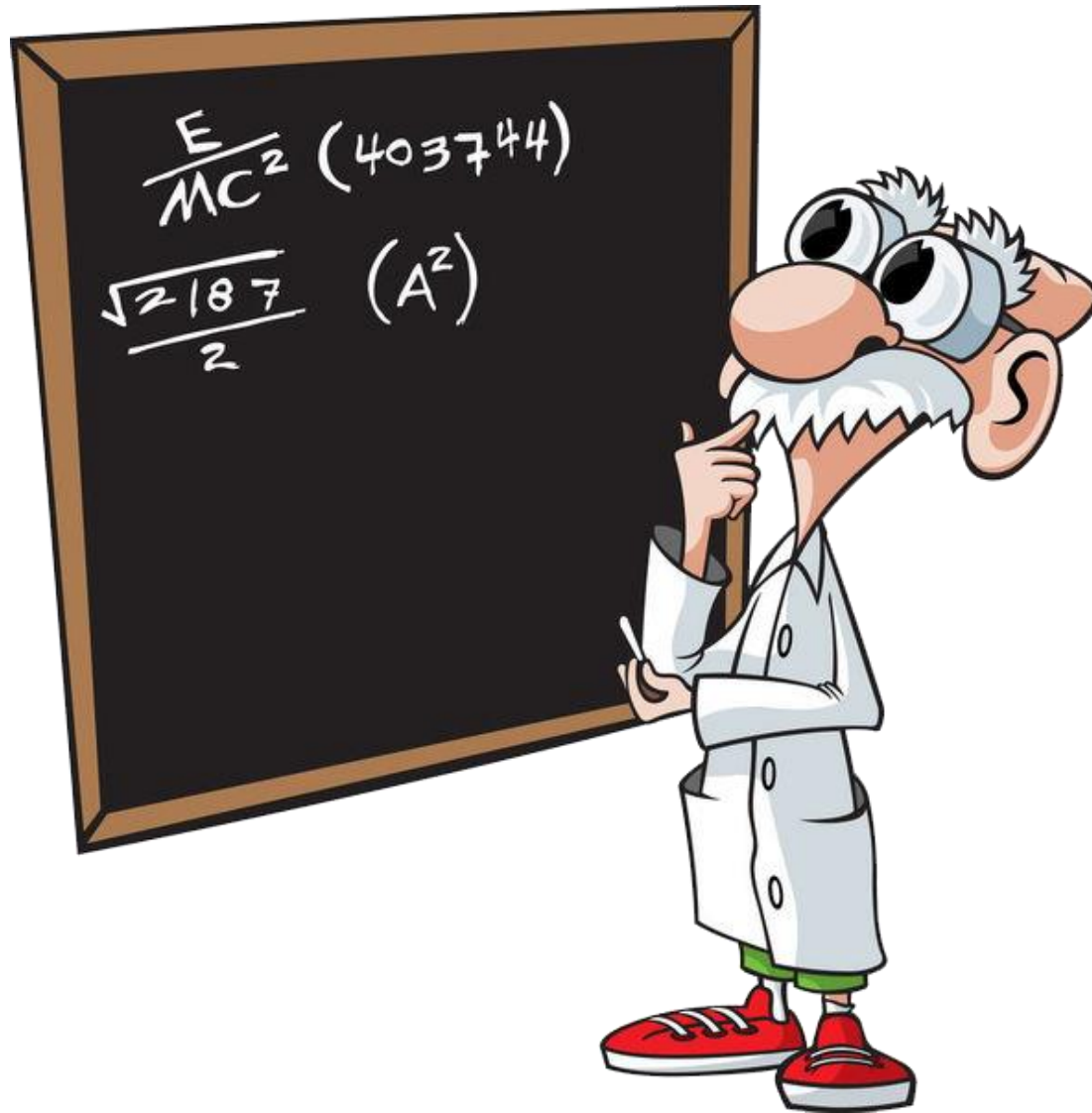
    // SQL statement to delete the row
    $sql_delete = "DELETE FROM administrator WHERE id=$delete_id";

    // Execute the query and check for errors
    if ($conn->query($sql_delete) === TRUE) {
        echo "Record deleted successfully<br>";
        // Refresh the page to show the updated table
        header("Location: " . $_SERVER['PHP_SELF']);
        exit();
    } else {
        echo "Error deleting record: " . $conn->error . "<br>";
    }
}
```

ID	First Name	Last Name	Email	Action
36	Jane	Smith	jane.smith@example.com	Delete

Exercise

How can you start creating a university database and a students table with columns for student ID, name, email, city, and age, insert data for two students into the table, and then fetch and display this data in an HTML table on a webpage?



Thank You